Czech Academy of Sciences–Kanazawa University Workshop 2025

Shiinoki Cultural Complex, Seminar Rooms A and B
Kanazawa City, Ishikawa Prefecture, Japan

November 02 – 03, 2025

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What is CAS–KU 2025?

The CAS–KU Workshop 2025 brings together researchers from the Czech Academy of Sciences, Kanazawa University, and partner institutions for a two-day interdisciplinary meeting in Kanazawa, Japan. The workshop aims to introduce the main research projects led by members of these institutions and to explore potential collaborations and future research initiatives. The program includes long, medium, and short talks, as well as discussions, fostering the exchange of ideas and strengthening scientific ties across institutions.

Acknowledgements

Date and Venue

The workshop is held on Sunday, November 02, and Monday, November 03, 2025, at seminar rooms A and B in Shiinoki Cultural Complex, Kanazawa City, Ishikawa Prefecture, Japan.

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Speakers

Afifah Maya IKNANINGRUM	Kanazawa University, Japan
Yoshiyuki KAGEI	Insitute of Science Tokyo, Japan
Martin KALOUSEK	Czech Academy of Sciences
Yuki KARASAWA	Kanazawa University, Japan
Hideo KOZONO	Waseda University, Japan
Ondřej KREML	Czech Academy of Sciences
Václav MACHA	Czech Academy of Sciences
Patrick van MEURS	Kanazawa University, Japan
Sahat Pandapotan NAINGGOLAN	Kanazawa University, Japan
Yuma NAKAMURA	Kanazawa University, Japan
Šárka NEČASOVÁ	Czech Academy of Sciences
Hirofumi NOTSU	Kanazawa University, Japan
Naohiko OGAWA	Kanazawa University, Japan
Oussama OUNISSI	Kanazawa University, Japan
Norbert POŽÁR	Kanazawa University, Japan
Kharisma Surya PUTRI	Kanazawa University, Japan
Koya SAKAKIBARA	Kanazawa University, Japan
Jan VALAŠEK	Czech Academy of Sciences

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Oussama Ounissi	Kanazawa University, Japan
Kharisma Surya Putri	Kanazawa University, Japan

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Day 1, 10:00 - 10:50, Nečasová

Weak solutions to a full compressible magnetohydrodynamic flow interacting with thermoelastic structure and singular limits

Kuntal Bhandari, Šárka Nečasová
Institute of Mathematics, Czech Academy of Sciences, Czech Republic

Bingkang Huang
University of Technology, Hefei, China

Sourav Mitra
Indian Institute of Technology Indore, India

Yadong Liu
Nanjing Normal University, China

This lecture is devoted to the problem interaction between a full compressible, electrically conducting fluid and a thermoelastic shell in a two-dimensional setting. The shell is modeled by linear thermoelasticity equations, and encompasses a time-dependent domain which is filled with a fluid described by full compressible (non-resistive) magnetohydrodynamic equations. The magnetohydrodynamic flow and the shell are fully coupled, resulting in a fluid-structure interaction problem that involves heat exchange. We establish the existence of weak solutions through *domain extension*, *operator splitting*, *decoupling*, *penalization of the interface condition*, and *appropriate limit passages*. It is a joint work with Kuntal Bhandari and Bingkang Huang, [1].

Moreover, we give a rigorous justification of the incompressible inviscid limit of the compressible fluid-structure interaction problem with a flat reference geometry, in the regime of low Mach number, high Reynolds number, and well-prepared initial data in the case of barotropic fluid without thermal or magnetic effect. It is a joint work with Sourav Mitra and Yadong Liu, [2].

References

[1] Kuntal Bhandari, Bingkang Huang,Šárka Nečasová *Weak solutions to a full compressible magnetohydrodynamic flow interacting with thermoelastic structure* , arXiv:2505.23539, 2025.

[2] Yadong Liu, Sourav Mitra, Šárka Nečasová *Weak solutions and singular limits for a compressible fluid-structure interaction problem with slip boundary conditions*, arXiv:2405.09908, 2025.

Day 1, 10:50 – 11:05, Iknaningrum

Non-axisymmetric Tornado-type Flow: Energy Transfer and Dynamics

Afifah Maya Iknaningrum
Graduate School of Natural Science and Technology, Kanazawa University, Japan

Pen-Yuan Hsu
Graduate School of Mathematical Science, The University of Tokyo, Japan

Tsuyoshi Yoneda
Graduate School of Economics, Hitotsubashi University, Japan

Hirofumi Notsu
Faculty of Mathematics and Physics, Kanazawa University, Japan

While tornado-type flows have been extensively studied in straight cylindrical domains (e.g. [1]), the influence of curvature on vortex dynamics remains less understood. Incorporate curvature to 3D numerical simulation in a non-axisymmetric curved cylindrical domain, we reveals specific vortex dynamics responsible for energy transfer [2]. Recent studies related to scale-local energy transfer [3] and observational studies of real tornado [4], supporting the credibility of our numerical approach.

References

[1] Hsu, P.Y., Notsu, H. and Yoneda, T., *A local analysis of the axisymmetric Navier-Stokes flow near a saddle point and no-slip flat boundary*, J. Fluid Mech., 729, pp. 444–459, 2016.

[2] Iknaningrum, A.M., Hsu, P.Y., Yoneda, T., and Notsu, H., *Energy Transfer Dynamics Generated by Non-Axisymmetric Tornado-Type Flows*, [Under review], 2025.

[3] Goto, S., Saito, Y. and Kawahara, G., *Hierarchy of antiparallel vortex tubes in spatially periodic turbulence at high Reynolds numbers*, Physical Review Fluids, 2, 064603, 2017.

[4] Bluestein, H.B., Thiem, K.J., Snyder, J.C. and Houser, J.B., *The Multiple-Vortex Structure of the El Reno, Oklahoma, Tornado on 31 May 2013*, Monthly Weather Review, 146(8), pp. 2483–250, 2018.

Day 1, 11:20 – 11:35, Karasawa

A Variable Time Step Lagrange-Galerkin Scheme with Second-Order Accuracy in Time for Convection-Diffusion Problems

Yuki karasawa
Kanazawa University, Japan

This presentation proposes a variable time step formulation of the Lagrange–Galerkin method with second-order accuracy in time for the convection-diffusion problems. By employing the method of characteristics, the scheme reduces numerical diffusion and relaxes CFL restrictions [1]. Stability is demonstrated using ideas from [2] and [3], and norm error estimates are derived from the stability results.

References

[1] K. Futai, N. Kolbe, H. Notsu, and T. Suzuki, *A mass-preserving two-step lagrange–galerkin scheme for convection-diffusion problems*, Journal of Scientific Computing, 92(2):37, 2022

[2] Liao, Hong-lin and Zhang, Zhimin, *Analysis of adaptive BDF2 scheme for diffusion equations*, Mathematics of Computation, 90, 1207–1226, 2021

[3] W. Wang, M. Mao, and Z. Wang, *Stability and error estimates for the variable step-size BDF2 method for linear and semilinear parabolic equations*, Advances in Computational Mathematics, 47(1):8, 2021

Day 1, 11:35 – 12:05, Požár

A rate-independent model of droplet evolution

Norbert Pozar
Kanazawa University, Japan

In this talk I will introduce a simplified model of a quasistatic droplet on a surface with contact angle hysteresis based on a rate-independent evolution of the one-phase Bernoulli free boundary problem. Taking advantage of two notions of weak solutions, energy-based and comparison-principle-based, we study the dynamic contact angle of moving contact lines and the geometry of de-pinning. We show that these two notions of solutions coincide in a star-shaped setting, where we show (almost) optimal regularity of the contact line and the convergence of a minimizing movements scheme. In a general setting, the notions differ essentially in how they handle jumps, but both are shown to satisfy a weak motion law. This talk is based on joint work with Inwon Kim (UCLA) and William M. Feldman (University of Utah), [1] [2] [3].

References

[1] W. M. Feldman, I. Kim, N. Požár, *An obstacle approach to rate independent droplet evolution*, arXiv:2410.06931

[2] W. M. Feldman, I. Kim, N. Požár, *On the geometry of rate independent droplet evolution*, arXiv:2310.03656

[3] W. M. Feldman, N. Požár, *On convex comparison for exterior Bernoulli problems with discontinuous anisotropy*, Interfaces Free Bound. 27 (2025), no. 3, 459–468.

Day 1, 14:00 – 14:30, van Meurs

3 topics on many-particle limits

Patrick van Meurs
Kanazawa University, Japan

I will introduce my main 3 research topics in applied analysis:

1. The many-particle limit of particle systems that are described by either the minimizer of an interacting particle energy (as the equilibrium state), or as the gradient flow thereof. The challenge is that the particle interaction is singular.

2. One particular model is a system of moving Coulomb charges, to which I consider added noise. Proving a many-particle limit is very ambitious; at the moment I try to build a well-defined solution concept for this system of singular SDEs and to figure out what kind of collisions can occur.

3. There is a lot of interest in proving that several types of PDEs ((non)linear heat equation with a reaction term, Navier-Stokes) emerge as the many-particle limit (= hydrodynamic limit) of rather simple stochastic particle systems (= simple cellular automata). I work on this too.

Day 1, 14:30 – 15:20, Kalousek

Analysis of bi-fluid systems

Martin Kalousek
Institute of Mathematics CAS, Czech republic

The talk is devoted to bi-fluid systems. After their introduction the attention will be focused on the existence analysis for two examples. Namely, we will briefly discuss the global in time existence of a weak solution to so called one velocity Baer-Nunziato system and the existence of a weak solution to a bi-fluid system in a bounded domain with variable boundary. The presented results come from papers [1, 2].

References

[1] Kalousek M., Mitra S., Nečasová Š., *The existence of a weak solution for a compressible multicomponent fluid structure interaction problem*, J. Math. Pures Appl., 184, 118-189, 2024.

[2] Kalousek M., Nečasová Š. *On existence of weak solutions to a Baer–Nunziato type system*, J. Differential Equations, 452, 2026.

Day 1, 15:35 – 15:50, Nakamura

On the memory of the twin vortex computer for an optimized cylinder

Yuma Nakamura
Kanazawa University, Japan

John Sebastian H. Simon
RICAM, Austria

Tomoyuki Kubota, Kohei Nakajima
The University of Tokyo, Japan

Hirofumi Notsu
Kanazawa University, Japan

In recent years, forecasting certain phenomena based on given robust data has been of interest among mathematicians and allied scientists. Our interest lies in the study of a method known as *Physical Reservoir Computing* which is derived from the framework of recurrent neural network. In particular, we consider the dynamics induced by the flow past a cylinder which was studied in [1]. In the said article the authors studied the effect of varying the so-called Reynolds number to the information processing capacity. They showed that the case where the twin-vortex — generated due to the periodically stable flow — appears, the physical reservoir computer gives a meaningful information processing and reaches peak capability when the Reynolds number is at the bifurcation point of stability. In this talk, we present an on going study on the effect of varying the shape of the obstacle in the channel by using shape optimization. The optimization problem is formulated so as to either maximize or minimize the L^2-norm of the curl of the velocity field based on the result of [1] indicating that the size of the vortex affects the computational capability of the system. With the use of traction method based on gradient descent of the vorticity functional, we numerically solve the shape optimization problem. The performance of the physical reservoir computer is then evaluated on the shapes generated from the optimization problems. We then present our early results which shows that the shapes generated by the minimization problem give a better performance compared to the maximized one. This is a joint work with John Sebastian H. Simon, Tomoyuki Kubota, Kohei Nakajima, and Hirofumi Notsu.

References

[1] K. Goto, K. Nakajima and H. Notsu. Twin vortex computer in fluid flow. *New Journal of Physics*, 23: 063051, 2021.

Day 1, 15:50 – 16:40, Macha

On compressible fluids with shear dependent viscosity

Václav Mácha
Institute of Mathematics, Czech Academy of Sciences

We present recent developments concerning the existence of solutions to the generalized Navier–Stokes system describing the flow of a compressible fluid with shear-dependent viscosity. Establishing existence in this setting is particularly delicate, as the equations do not permit deriving sufficiently strong compactness properties to apply the standard Galerkin method for obtaining weak solutions. Instead, we discuss several results regarding measure-valued and strong solutions to the aforementioned system.

During the talk, I will present results obtained in the collaboration with M. Kalousek, P. Ludvík, Š. Nečasová, and J. Slavík.

Day 1, 16:55 – 17:45, Kreml

On dissipative turbulent solutions to the compressible anisotropic Navier-Stokes equations in unbounded domains

Ondřej Kreml
Institute of Mathematics, Czech Academy of Sciences, Czechia

Inspired by Abbatiello, Feireisl and Novotný [1], we prove the global existence of dissipative turbulent solution for the compressible Navier-Stokes equations with anisotropic viscous stress tensor on unbounded domain. Our work complements the result of Bresch and Jabin [2], where the authors used the new compactness method to prove the existence of a weak solution to the same system in the 3D torus. By working with larger class of dissipative turbulent solutions, we are able to relax assumptions on the anisotropic tensor coefficients and the pressure law coefficient and we establish the existence result on a large class of unbounded domains, which is more conform to physical context. We also prove the weak-strong uniqueness property of acquired dissipative turbulent solutions.

This is a joint work with Šárka Nečasová and Tong Tang.

References

[1] A. Abbatiello, E. Feireisl, and A. Novotný, *Generalized solutions to models of compressible viscous fluids*, Discrete Contin. Dyn. Syst., 41, 1–28, 2021.

[2] D. Bresch and P. E. Jabin, *Global existence of weak solutions for compressible Navier-Stokes equations: thermodynamically unstable pressure and anisotropic viscous stress tensor*, Ann. of Math., 188, 577–684, 2018.

Day 2, 09:30 – 10:20, Kozono

Equilibrium state of the 3D MHD equations with an arbitrary geometry

Hideo Kozono
Waseda University & Tohoku University, Japan

Senjo Shimizu
Kyoto Univresity, Japan

Taku Yanagisawa
Nara Women's Univresity, Japan

In the 3-dimensional bounded domain Ω with smooth boundary $\partial\Omega$, we consider stability problem of solutions to the MHD equations for $U = (u, B)$ denoting the velocity and magnetic field, respectively. Introducing the harmonic vector field

$$X_{har}(\Omega) = \{h \in C^\infty(\bar{\Omega}); \operatorname{div} h = 0, \operatorname{rot} h = 0, h \cdot \nu|_{\partial\Omega} = 0\}$$

with ν denoting the unit outer normal to $\partial\Omega$, we first show that $U_* = (0, B_*)$ with $B_* \in X_{har}(\Omega)$ gives an equilibrium state. Next, we prove that if B_* is small in $L^1(\Omega)$ and if the initial disturbance U_0 is sufficiently small in $L^3(\Omega)$ with $B_0 - B_*$ perpendicular to $X_{har}(\Omega)$, then U_* is exponentially stable. This is based on papers [1, 2].

References

[1] Kozono, H., Yanagisawa, T., *Variational inequality for vector fields and the Helmholtz–Weyl decomposition in bounded domains.* Inidiana Univ. Math, J. **58** (2009), 1853–1920.

[2] Kozono, H., Shimizu, S., Yanagisawa, T., *Stability of harmonic vector fields as an equilibrium of the 3D MHD equations in bounded domains with arbitrary geometry.* Math. Ann. **393** (2025), 923–952.

Day 2, 10:20 – 10:35, Nainggolan

Parameter Identification in Elliptic PDEs Using the Coupled Complex Boundary Method with Tikhonov Regularization

SAHAT PANDAPOTAN NAINGGOLAN
Division of Mathematical and Physical Sciences, Kanazawa University, Japan

JULIUS FERGY TIONGSON RABAGO, HIROFUMI NOTSU
Faculty of Mathematics and Physics, Kanazawa University, Japan

This paper investigates an inverse coefficient problem for a convection–reaction–diffusion equation, where the objective is to reconstruct a spatially varying diffusion coefficient from limited boundary measurements. Both Dirichlet and Neumann data are incorporated into a unified framework using the Coupled Complex Boundary Method (CCBM) [1, 3, 5]. In this setting, the Cauchy data are encoded as a complex Robin-type boundary condition, allowing the boundary fitting condition to be transferred into the domain interior. To stabilize the reconstruction, Tikhonov regularization [2, 4] is employed, ensuring the well-posedness of the variational formulation. The forward and inverse problems are discretized using the finite element method and implemented in FreeFEM++. Numerical experiments demonstrate the feasibility and robustness of the proposed approach, showing accurate recovery of the diffusion coefficient even in the presence of measurement noise.

References

[1] X. L. Cheng, R. F. Gong, W. Han, and X. Zheng, A novel coupled complex boundary method for solving inverse source problems, *Inverse Problems*, 30 (2014), p. 055002.

[2] H. W. Engl, M. Hanke, and A. Neubauer, *Regularization of Inverse Problems*, Springer, 1996.

[3] R. F. Gong, X. L. Cheng, and W. Han, A coupled complex boundary method for an inverse conductivity problem with one measurement, *Applicable Analysis*, 96 (2017), pp. 869–885.

[4] A. N. Tikhonov and V. Y. Arsenin, *Solutions of Ill-posed Problems*, Wiley, New York, 1977.

[5] X. Zheng, X. Cheng, and R. Gong, A coupled complex boundary method for parameter identification in elliptic problems, *International Journal of Computer Mathematics*, 97 (2020), pp. 998–1015.

Day 2, 10:35 – 10:50, Ounissi

Fracture phase field model with unilateral contact condition: energy dissipation identity and finite element simulations

Oussama Ounissi, Masato Kimura
Kanazawa University, Japan

Md Mamun Miah
Khulna University of Engineering and Technology, Bangladesh

Sayahdin Alfat
Halu Oleo University, Indonesia

The fracture phase field model (PFM) is widely used to model fracture mechanics, but in certain scenarios the standard PFM fails to reproduce realistic crack propagation because it does not consider the unilateral contact (non-penetration) conditions. In this talk, we present a variational formulation of the phase-field model that incorporates unilateral contact, and prove an energy-dissipation identity for both the standard and with unilateral contact formulations. We will also illustrate the effect of the contact condition through numerical examples.

Day 2, 11:05 – 11:20, Putri

Lagrangian–Galerkin Moving Mesh Method

Kharisma Surya Putri, Tatsuki Mizuochi
Division of Mathematical and Physical Sciences, Kanazawa University, Japan

Niklas Kolbe
Institute of Geometry and Applied Mathematics, RWTH Aachen University, Germany

Hirofumi Notsu
Faculty of Mathematics and Physics, Kanazawa University, Japan

In this research, our goal is to accurately capture high-concentration phenomena that lead to sharp spike patterns, such as those observed in cancer cell aggregation. These problems are typically modeled by convection-dominated PDEs, which require careful domain discretization to achieve high-resolution results. In recent years, adaptive mesh methods and optimal transport techniques have gained popularity as effective approaches for handling such challenges.

In this talk, based on joint work with T. Mizuochi, N. Kolbe, and H. Notsu (see [1]), we introduce the **Lagrangian–Galerkin Moving Mesh Method (LGMM)** as a low-cost computational alternative for handling high-concentration phenomena. Here, LGMM extends the mass-conservative Lagrange–Galerkin (LG) framework, a powerful method for convection-dominated problems, into a moving mesh setting. In this approach, mesh nodes are dynamically moved to follow the underlying flow, while maintaining the mass conservation property inherent in fixed-mesh LG methods.

By carefully selecting time increments, we can prevent mesh overlap and degeneration, ensuring a stable and accurate solution process. Moreover, for linear elements, we establish an optimal error estimate in the $\ell^\infty(L^2) \cap \ell^2(H_0^1)$ norms. Through numerical experiments, we validate our theoretical findings and highlight several advantages of our moving mesh approach compared to the standard fixed-mesh Lagrangian-Galerkin method.

References

[1] Putri, K.S., Mizuochi, T., Kolbe, N., Notsu, H., *Error Estimates for First- and Second-Order Lagrange–Galerkin Moving Mesh Schemes for the One-Dimensional Convection–Diffusion Equation.* Journal of Scientific Computing 101, 2024.

Day 2, 11:20 – 12:10, Valašek

Numerical modelling of human phonation process

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Institute of Mathematics, Czech Academy of Sciences, Žitná 25, Praha,CZ

Petr Sváček[2]
FME, Czech Technical University in Prague, Karlovo nám 13, Praha 2, CZ

This contribution focuses on the fluid–structure–acoustic interaction (FSAI) problem with particular emphasis on its application to human phonation. The process of phonation represents a highly complex interaction between three physical fields: the vibrating elastic structure of the vocal folds, the unsteady airflow, and the resulting acoustics, together with their mutual couplings. A general mathematical model of the FSAI problem is introduced, followed by its simplifications suitable for numerical treatment as e.g. approximation of the FSAI problem by a fluid–structure interaction (FSI) problem with a forward-coupled aeroacoustic model.

Particular attention is devoted to the mutual contact of the vocal folds, which regularly occurs during healthy phonation and substantially increases the complexity of the problem. A contact treatment compatible with the chosen numerical framework (finite element method) is described in [1].

The aeroacoustic problem is modelled by a aeroacoustic analogies (perturbation approach) having advantage of computationally feasible alternative to the full compressible Navier–Stokes equations. Two mathematical formulations — the Lighthill acoustic analogy and the Aeroacoustic Wave Equation (AWE) — are presented and applied, see [2].

Finally, results of numerical simulations of flow-induced vocal fold vibrations and sound propagation through the human vocal tract are shown for the phonation of the vowel [u:], see [3].

References

[1] P. Sváček, J. Horáček, FE numerical simulation of incompressible airflow in the glottal channel periodically closed by self-sustained vocal folds vibration, J. Comput. Appl. Math. 393 (2021) 113529 DOI 10.1016/j.cam.2021.113529

[2] Falk, S., Kniesburges, S., Schoder, S., etal.: 3D-FV-FE aeroacoustic larynx model for investigation of functional based voice disorders. Front. Physiol. 12 (2021) 616985 DOI 10.3389/fphys.2021.616985

[3] Valášek, J., Sváček, P.: Aeroacoustic simulation of human phonation based on the flow-induced vocal fold vibrations including their contact. Adv. Eng. Softw. 194 (2024) DOI 10.1016/j.advengsoft.2024.103652

Day 2, 14:00 – 14:30, Sakakibara

Well-posedness of the Langmuir film model

Yoichiro Mori
University of Pennsylvania, USA

Koya Sakakibara
Kanazawa University, Japan

Shinya Okabe
Tohoku University, Japan

A Langmuir film (or Langmuir monolayer) is a single molecular layer of amphiphilic molecules spread at the air–water interface, exhibiting two-dimensional fluid behavior coupled with the subphase hydrodynamics beneath it [3, 2]. We investigate the inviscid Langmuir layer–Stokes subfluid (ILLSS) model [1], which couples a two-dimensional, incompressible, inviscid film on $\partial B = \mathbb{R}^2$ with a three-dimensional Stokes flow in the subfluid $B = \mathbb{R}^2 \times (-\infty, 0)$. The governing system is given by

$$\begin{cases} -\triangle \boldsymbol{v} + \nabla q = \boldsymbol{0}, \quad \nabla \cdot \boldsymbol{v} = 0, & \text{in } B, \\ v_3 = 0, \quad \boldsymbol{v}_{\parallel}|_{\partial B} = \boldsymbol{u}, & \text{on } \partial B, \\ \nabla p = -\partial_{x_3} \boldsymbol{v}_{\parallel}|_{\partial B}, \quad \nabla \cdot \boldsymbol{u} = 0, & \text{on } \partial B \setminus \Gamma, \\ [\![\boldsymbol{u}]\!] = \boldsymbol{0}, \quad [\![p]\!] = \kappa, \quad U = \boldsymbol{u} \cdot \boldsymbol{\nu} & \text{on } \Gamma, \end{cases}$$

where $\Gamma \subset \partial B$ denotes the moving closed interface, κ its curvature, and U its normal velocity.

In this talk, we first derive the boundary integral equation (BIE) governing the evolution of Γ and establish a curve-shortening identity. Next, we prove the local-in-time existence of solutions to the BIE based on maximal L^2-regularity in H^1. Furthermore, by obtaining a regularity result, we show that the ILLSS model is equivalent to the BIE formulation. Finally, we present several numerical experiments illustrating the dynamics.

References

[1] J. C. Alexander, A. J. Bernoff, E. K. Mann, J. A. Mann, Jr., J. R. Wintersmith, and L. Zou, *Domain relaxation in Langmuir films*, J. Fluid Mech., 571, 191–219, 2017.

[2] K. B. Blodgett, *Films built by depositing successive monomolecular layers on a solid surface*, J. Am. Chem. Soc., 57, 1007–1022, 1935.

[3] I. Langmuir, *The constitution and fundamental properties of solids and liquids, II. Liquids.*, J. Am. Chem. Soc., 39, 1848–1906, 1917.

Day 2, 14:30 – 15:20, Kagei

Stability of bifurcating patterns in viscous compressible fluids

Yoshiyuki Kagei
Institute of Science Tokyo, Japan

In systems of equations for motion of viscous fluids, solutions often exhibit various interesting spatio-temporal periodic patterns. In this talk, we mainly consider the stability of the bifurcating spatially periodic patterns in the compressible Navier-Stokes system for the Couette-Taylor flows.

Day 2, 15:35 – 15:50, Ogawa

Kernel-based numerical schemes for geometric flows

Naohiko Ogawa, Koya Sakakibara
Kanazawa University, Japan

Yuka Hashimoto
NTT Network Technology Laboratories, Japan

Geometric flows often involve challenging numerical issues due to evolving geometries. In this talk, we study these problems within the framework of Reproducing Kernel Hilbert Space, where curves can be represented and evolved through kernel functions. As an application, we consider the curve shortening flow and verify the effectiveness of our proposed scheme through numerical experiments. This is a joint work with Koya Sakakibara (Kanazawa University) and Yuka Hashimoto (NTT Network Technology Laboratories).

Day 2, 15:50 – 16:40, Notsu

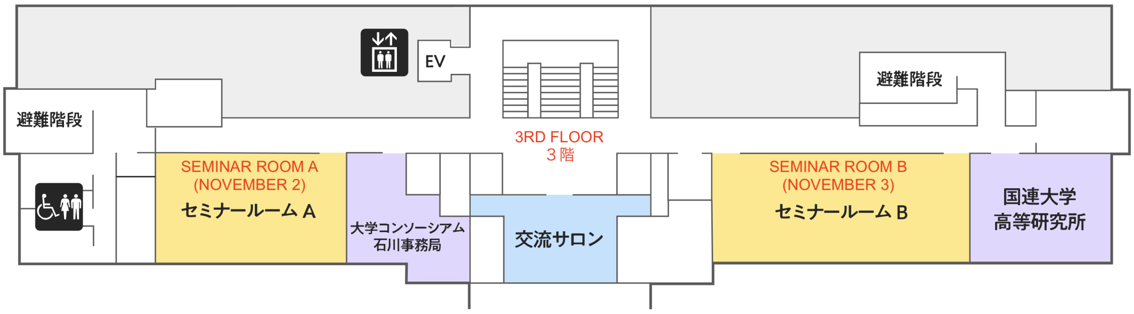
Spatially adaptive stabilized Lagrange–Galerkin schemes for two-fluid flow and fluid-structure interaction problems

Hirofumi Notsu, Kouta Futai, Yuki Karasawa
Kanazawa University, Japan

Lagrange–Galerkin (LG) schemes combined with pressure stabilization (PS), an adaptive mesh refinement (AMR) technique, and a level set method (LSM) are presented for two-fluid flow (TFF) and fluid-structure interaction (FSI) problems. LG schemes are robust for convection-dominated problems and provide symmetric approximations of the nonlinear material derivatives without artificial parameters, as they are based on the method of characteristics. An advantage of LG schemes, at least for problems with a linear convection term, is that no CFL condition is required; thus, the time increment can be chosen independently of the mesh size. LG schemes are therefore potentially compatible with AMR techniques. With the aid of PS, our schemes employ the simplest piecewise linear (P1) conforming finite element for all the unknowns—the velocity, pressure, and level set (signed distance) function. Consequently, the degrees of freedom can be easily controlled, and the schemes are particularly effective in three dimensions. As a foundation of this study, we introduce a second-order-in-time adaptive stabilized LG scheme with a mass-preserving approximation of the material derivative for the incompressible Oseen (linearized Navier–Stokes) equations, and we prove its theoretical properties, namely stability and error estimates. The proposed schemes for TFF and FSI problems are then presented. In these schemes, integrals over moving domains and boundaries are transformed into integrals over the entire domain, independent of time, by using approximate Heaviside step and Dirac delta functions. The AMR technique refines meshes near moving boundaries to accurately capture interface phenomena. Numerical results in two and three dimensions demonstrate the effectiveness of the proposed adaptive stabilized LG schemes.

Venue

The venue of the workshop/conference is the seminar rooms A & B, located on the third floor of the Shiinoki Cultural Complex.


EV
避難階段
3RD FLOOR
3 階
避難階段
SEMINAR ROOM A
(NOVEMBER 2)
セミナールーム A
大学コンソーシアム
石川事務局
交流サロン
SEMINAR ROOM B
(NOVEMBER 3)
セミナールーム B
国連大学
高等研究所


SEMINAR ROOM A
(November 2)

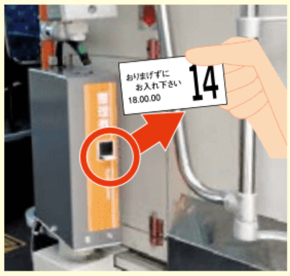

SEMINAR ROOM B
(November 3)

How to Ride Local-Line Buses

Enter the bus from the rear door.
Pay the fare when you get off.

Be sure to confirm the destination shown on the front and the side of the bus.

1 Take a numbered ticket (called "Sei-ri-ken") from the machine when getting on the bus.
Please note that numbered tickets ("Seiri-ken") are not issued on Kanazawa Loop Buses and Kanazawa Flat Buses. Simply get on those buses without taking a ticket.



Pay the bus fare when getting off. This numbered ticket shows the bus stop where you got on, and is required to pay the fare.
*On Kanazawa Flat Buses, pay the fare when you get on.

2 After your stop is announced and shown on the display at the front of the bus, press the Request Stop button to let the driver know that you will get off at the next stop.
The button will light up.

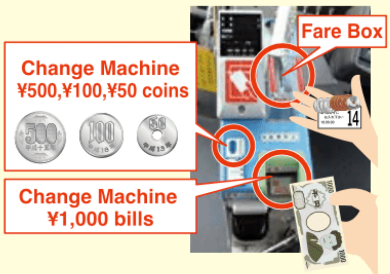
3 Confirm the fare shown on the display.

The child fare (for elementary school students ages 6-12) is half the adult fare (rounded up to the nearest ¥10).

4 When getting off, put both the fare and the numbered ticket in the fare box next to the driver.

Since change is not given, please use the change machine to put the exact fare in the fare box.

¥10,000, ¥5,000, and ¥2,000 bills cannot be changed.



Kanazawa Local Tourist Map

Strolling Major Sightseeing Spots by Kanazawa Loop Bus

Kanazawa Loop Bus (Left Loop)

Kanazawa Loop Bus (Right Loop)

Kanazawa Station

Get on Loop Bus at East Gate Bus Stop #7 — 0 0

Omicho Market

In front of 1 14

Higashi Chaya District (Geisha District)

Within 5 min from 10 4

Nagamachi Samurai District

Within 5 min from 3 12

Loop Bus Runs

Every 15 min

Adult: ¥210
Child: ¥110
(For one ride)

Kazue-machi Chaya Ditrict (Geisha District)

Within 5 min from 10 4

Nishi Chaya District (Geisha District)

Within 5 min from 5 10

Kanazawa Castle Park

Within 5 min from 9 6

21st Century Museum of Contemporary Art, Kanazawa

In front of 8 7

Kenrokuen Garden

Within 5 min from 9 6

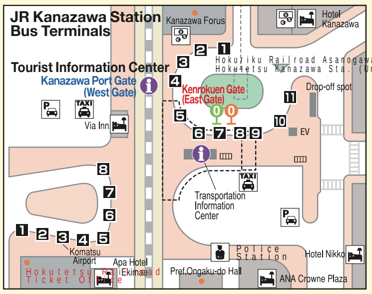
Bus Information

KANAZAWA LOOP BUS

Hours of operation:
Left Loop 8:30
Right Loop 8:35

● One Day Pass:
Adult: ¥800 Child: ¥400
Bus pass for Loop buses and regular-route buses in the central area

JR Kanazawa Station Bus Terminals



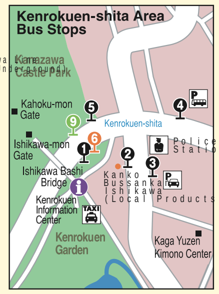
JR Kanazawa Sta. Kenrokuen Gate (East Gate) Bus Terminal

Kanazawa Loop Bus	7 0 0
Musashigatsuji·Omicho Market/Korinbo	3 7 - 11
Kenrokuen-shita·Kanazawa Castle Park	6 - 6
Hashiba-cho	6 7

JR Kanazawa Sta. Kanazawa Port Gate (West Gate) Bus Terminal

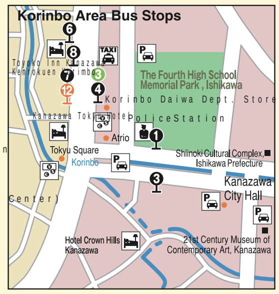
Komatsu Airport	3

Kenrokuen-shita Area Bus Stops



Kanazawa Loop Bus	9 6
Kanazawa Station/Korinbo Musashigatsuji·Omicho Market	1
Kodatsuno	3
Kanazawa Station/Musashigatsuji·Omicho Market/Hashiba-cho	2

Korinbo Area Bus Stops



Kanazawa Loop Bus	3 12
Kenrokuen-shita·Kanazawa Castle Park	1
Katamachi / Hirokoji	4
Musashigatsuji·Omicho Market/ Kanazawa Station	7

Japan Rail Pass is NOT available on Hokutetsu Bus.

Tourist Information Kanazawa Goodwill Guide Network (KGGN)

Kanazawa Castle Park Information Booth

- Guided tours of Kenrokuen Garden and Kanazawa Castle Park available free of charge. No reservation required.
- Available every day from 9:30 to 15:30 (except for year-end and New Year's holidays)

Emergency (Accidents/Natural Disaster)

In Case of Emergency, please visit the webpage from the QR code beside.

Guide for when you are feeling ill
Safety Tips
Traffic Information

Inquiry

Kanazawa Station Tourist Information Center : Phone +81-(0)76-232-6200
Kanazawa City Tourism Association : Phone +81-(0)76-232-5555
Tourism Policy Section of Kanazawa City Hall : Phone +81-(0)76-220-2194

VISIT KANAZAWA

@kanazawa_kankou

Visit Kanazawa

2024.4